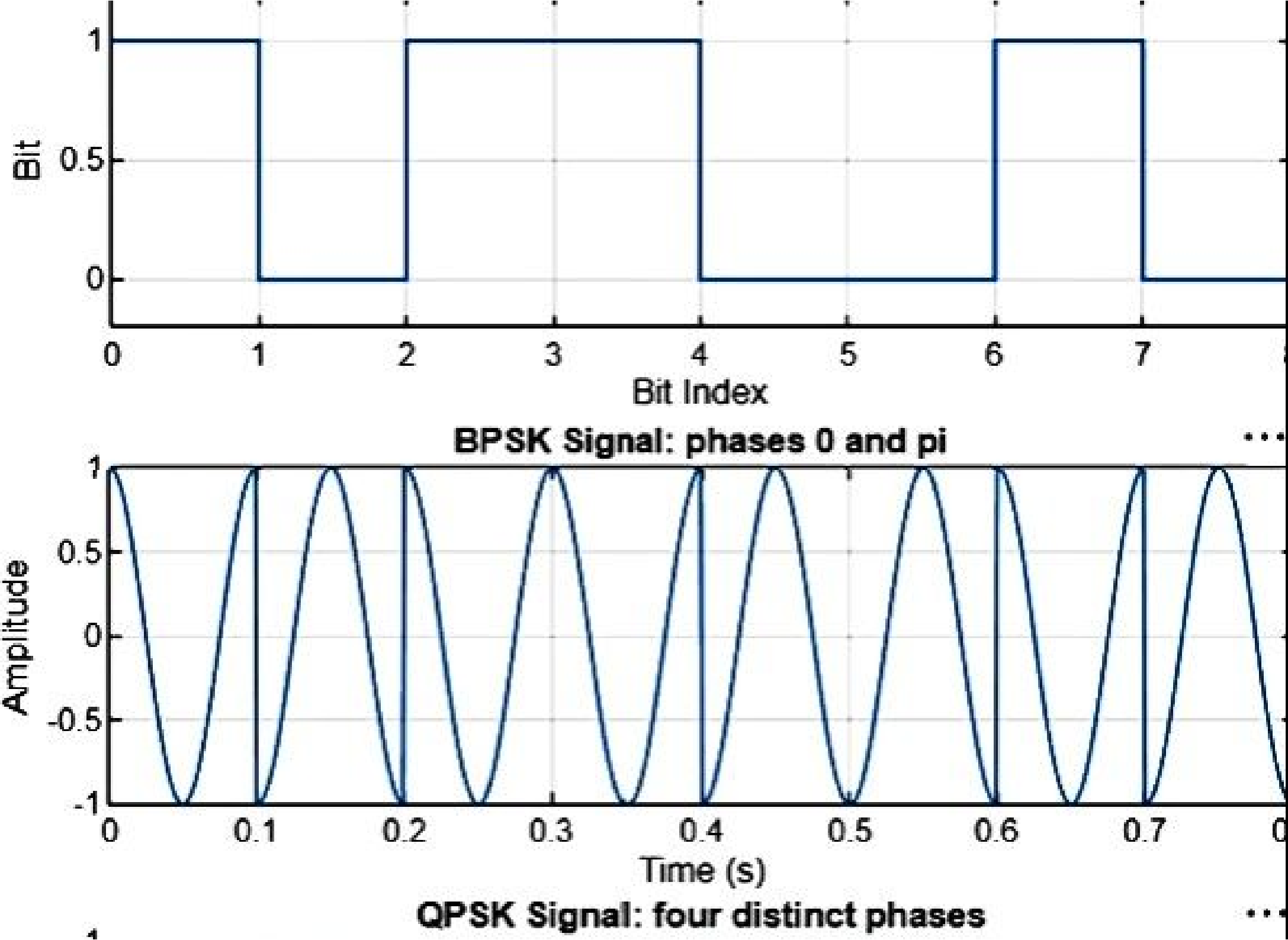
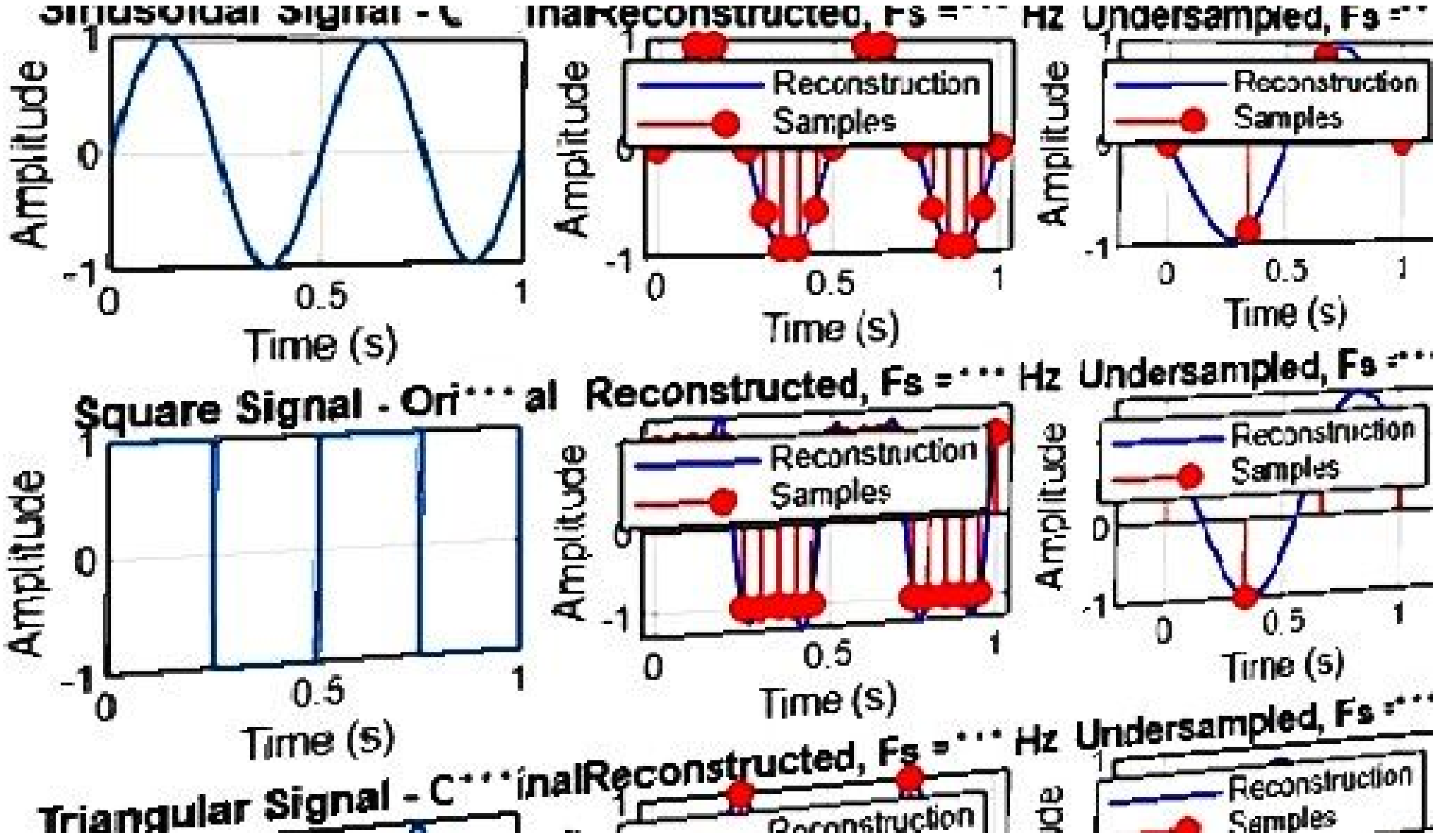
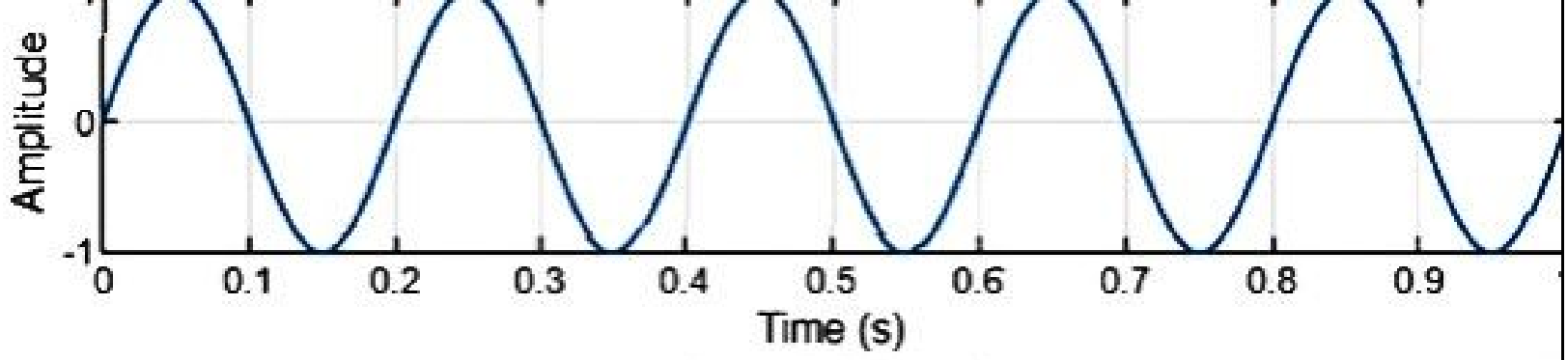
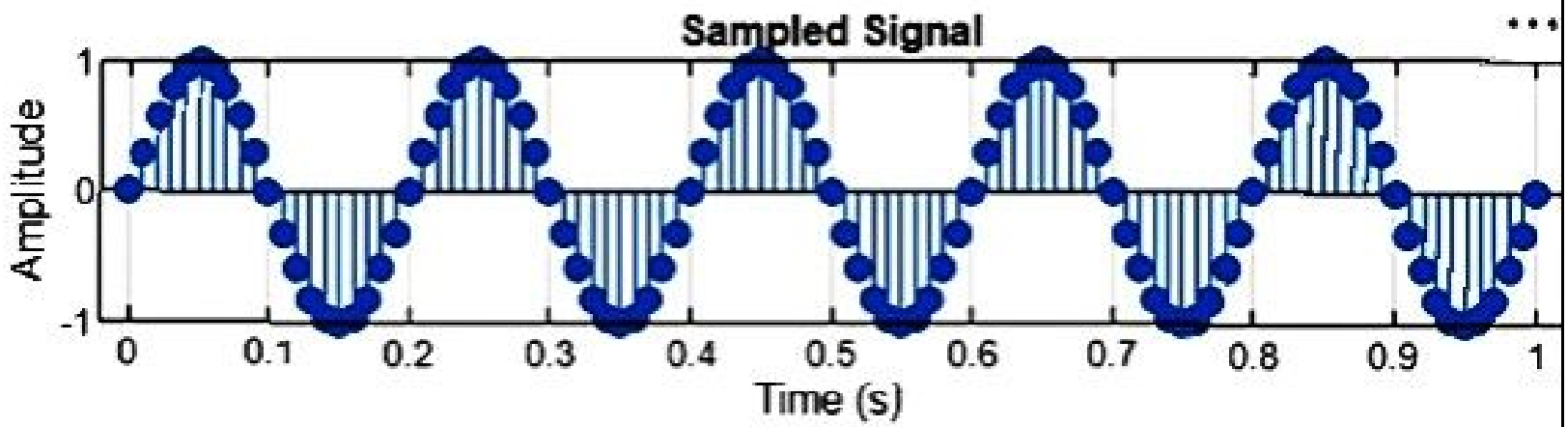
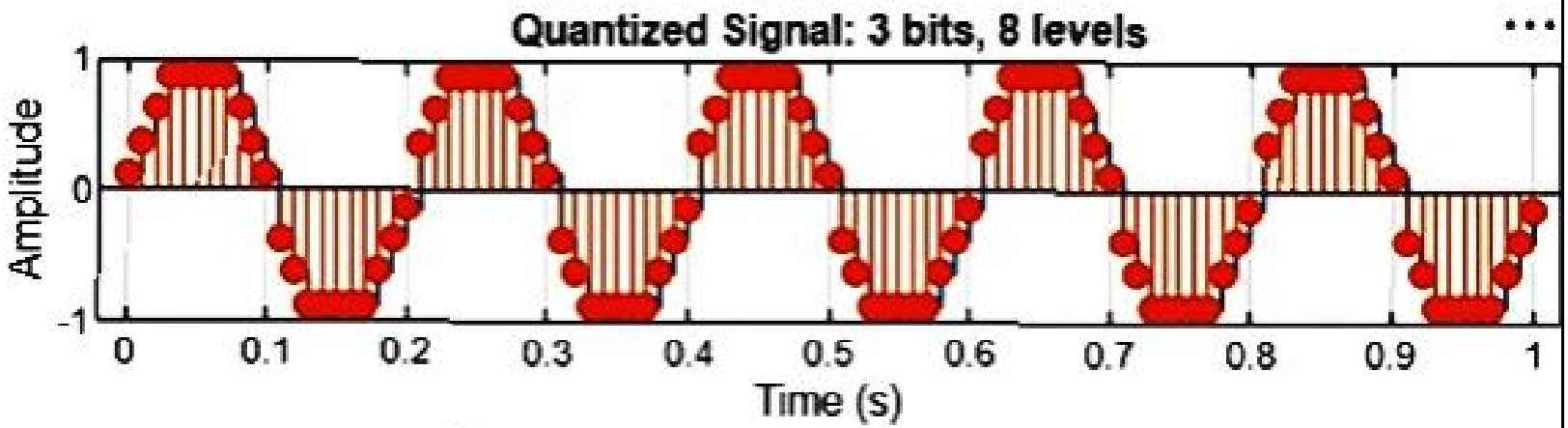
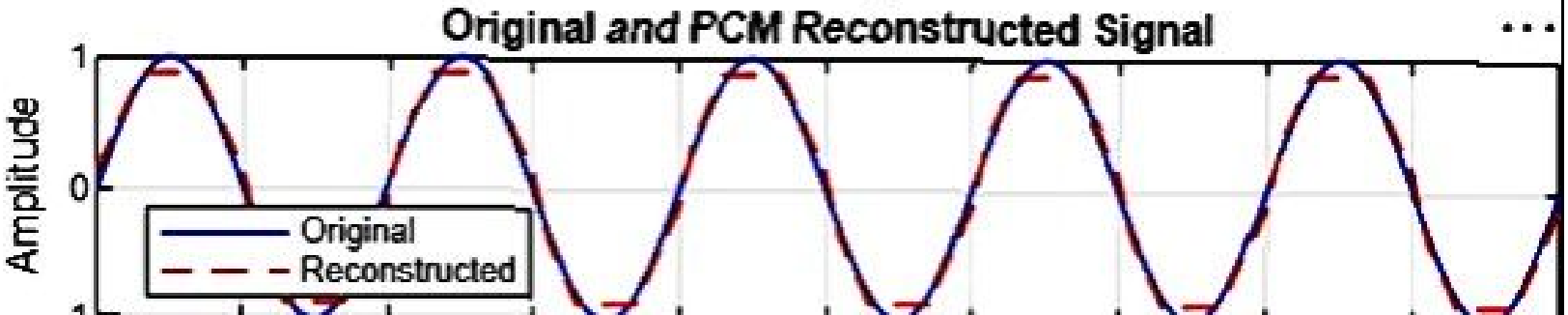

Bit
1
0.5
0
0
1
2
3
4
5
6
7
Bit Index
BPSK Signal: phases 0 and pi
Amplitude
1
0.5
0
-0.5
-1
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
Time (s)
QPSK Signal: four distinct phases


Reconstruction
Samples
Amplitude
Time (s)
0
0.5
1
-1


Amplitude
1
0
-1
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
Time (s)


Sampled Signal
Amplitude
1
0
-1
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
Time (s)


Quantized Signal: 3 bits, 8 levels
Amplitude
1
0
-1
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
Time (s)


Original and PCM Reconstructed Signal
Amplitude
1
0
Original
Reconstructed